

When does a rectangle fit into an n -dimensional box?

An explicit criterion with a formal proof

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Abstract

We give a necessary and sufficient condition for an $A \times B$ rectangle to fit, after an arbitrary rigid motion, into an n -dimensional rectangular box with edges $p_1, \dots, p_n$, for every $n \geq 2$. The condition is a finite disjunction: there must be two axes $k \neq l$ of the box and a partition of the remaining axes into sets S and T such that the rectangle with sides $\sqrt{A^2 - \|p_S\|^2}_+$ and $\sqrt{B^2 - \|p_T\|^2}_+$ must fit into the $p_k \times p_l$ rectangle, which is decided by Carver’s planar criterion. Geometrically, each side of the rectangle first runs along the diagonal of a coordinate subspace and the remainders are placed in one coordinate plane. This settles the case $k = 2$ of the general fitting problem for a k -dimensional box in a d -dimensional box stated by Jerrard and Wetzel in 2004; for $n = 3$ and the unit cube it reproduces their Theorem 5. The proof reduces the problem to the widths of the rectangle along the axes and studies a minimizer of the total “mixing” of the two sides. The theorem has been verified in the Lean 4 proof assistant with the Mathlib library. The criterion is constructive, and in Python it decides fitting in 6–200 microseconds for $n = 3, \dots, 6$.

1 Introduction

When does a box fit into another box? For a segment of length A and a box with edges $p_1, \dots, p_n$ the answer is immediate: $A^2 \leq p_1^2 + \dots + p_n^2$. Jerrard and Wetzel [2, p. 22] describe the general question — find necessary and sufficient conditions for a k -dimensional box to fit in a d -dimensional box — as “a general unsolved fitting problem for boxes in higher dimensions” and note that for $k = 1$ the matter is trivial. The first nontrivial case is a rectangle, $k = 2$. In the plane ($d = 2$) it was settled by Carver [1] (see also Wetzel [7]). For the unit cube ($d = 3$) the largest square has side $3\sqrt{2}/4$ (Nieuwland, published by van Swinden [6]; the problem of Prince Rupert), and Jerrard and Wetzel [2, Theorem 5] found the largest $L \times \lambda L$ rectangle in the unit cube. Explicit necessary and sufficient conditions of this kind are known for a few pairs of planar figures: a triangle in a triangle (Post [5]) and a rectangle in a triangle (Wetzel [8]); like the criterion below, they are finite disjunctions of explicit checks, one for each candidate position. The analogous problem for a three-dimensional box in a three-dimensional box was solved in [3]. Until now, an exact answer for a rectangle was known only in the plane (Carver) and for the unit cube (Jerrard and Wetzel).

In this paper we give such a criterion for every $n \geq 2$ (Theorem 2.1). It is a disjunction of $\frac{1}{2}n(n-1)2^{n-2}$ explicit checks, each of which is Carver’s planar test. The theorem says that an optimal position has a simple shape: each side of the rectangle is split into a part that runs along the diagonal of a coordinate subspace (spanned by the axes S for the side A and by the axes T for the side B) and a remainder lying in one coordinate plane (k, l) , where the two remainders are placed as a planar rectangle in the $p_k \times p_l$ rectangle.

The proof (Sections 3–7) is elementary. After the usual reduction to the widths of the rectangle along the axes (Section 4) the problem becomes a system in two unit vectors u, v with a single orthogonality condition. We describe the admissible configurations by two numbers (L, D) lying under a “tent” and take a configuration with the smallest D , the total mixing of the sides. Three lemmas then show that such a configuration either has no mixed axes or has exactly two, and in both cases the criterion holds.

The whole theorem has been checked in the Lean 4 proof assistant with the Mathlib library (Section 9). The formal statement uses only the definition of a box in $\mathbb{R}^n$ and of fitting by an isometry; the proof uses no axioms beyond the standard ones. The criterion is constructive: when the rectangle fits it gives an explicit position (Section 3), and a Python implementation is two to five orders of magnitude faster than a numerical search over rotations, which in addition often fails to find an existing position near the boundary (Section 10).

2 The theorem

2.1 Notation and Carver’s criterion

A *box* is $P = [0, p_1] \times \cdots \times [0, p_n] \subset \mathbb{R}^n$, $p_i \geq 0$, $n \geq 2$. An $A \times B$ *rectangle*, $A, B \geq 0$, is the set $\{x_0 + sAu + rBv : s, r \in [0, 1]\}$ for some $x_0 \in \mathbb{R}^n$ and some orthonormal pair $u, v \in \mathbb{R}^n$; it *fits* into P if x_0, u, v can be chosen so that it lies in P . Reflections are allowed (they change nothing: a rectangle is mirror-symmetric). For $S \subseteq [n] = \{1, \dots, n\}$ let $\|p_S\|^2 = \sum_{i \in S} p_i^2$ ($\|p_\emptyset\| = 0$), and $x_+ = \max(x, 0)$; we write $\sqrt{x}_+$ for $\sqrt{x_+}$.

Carver’s criterion [1] in the form of Wetzel [7, Theorem 1]. Let $X \geq Y \geq 0$ be the sides of a planar rectangle (the container) and $a \geq b \geq 0$ the sides of another. The $a \times b$ rectangle fits into the $X \times Y$ one if and only if

(i) $a \leq X$ and $b \leq Y$, or

(ii) $a > X$, $b \leq Y$ and $\left(\frac{X+Y}{a+b}\right)^2 + \left(\frac{X-Y}{a-b}\right)^2 \geq 2$.

We write $\text{Carver}(a, b; X, Y)$ for this condition after sorting both pairs in decreasing order. In case (ii) we have $a > X \geq Y \geq b$, so there is no division by zero. For $b = 0$ condition (ii) becomes $a^2 \leq X^2 + Y^2$, the criterion for a segment. We only use the monotonicity of the condition: if an $a \times b$ rectangle fits into $X \times Y$, then so does any $a_1 \times b_1$ with $a_1 \leq a$, $b_1 \leq b$, into any $X_1 \times Y_1$ with $X_1 \geq X$, $Y_1 \geq Y$.

2.2 Statement

Theorem 2.1. *Let $n \geq 2$, $p \in \mathbb{R}_{\geq 0}^n$ and $A, B \geq 0$. The $A \times B$ rectangle fits into the box p if and only if*

(II) *there are axes $k \neq l$ and a partition $S \sqcup T = [n] \setminus \{k, l\}$ such that*

$$\text{Carver}(a', b'; p_k, p_l), \quad a' = \sqrt{A^2 - \|p_S\|^2}_+, \quad b' = \sqrt{B^2 - \|p_T\|^2}_+.$$

The order $a' \geq b'$ is not guaranteed even when $A \geq B$ (for $A = 10$, $B = 9$, $\|p_S\| = 9$, $T = \emptyset$ we get $a' = \sqrt{19} < b' = 9$); this is why Carver sorts its arguments. Condition (II) is invariant under swapping k and l , so it consists of $\frac{1}{2}n(n-1)2^{n-2}$ checks: 1, 6, 24, 80, 240 for $n = 2, \dots, 6$. For $n = 2$ it is Carver’s criterion itself.

The meaning of (II) is shown by the construction in Section 3: the side A runs along the diagonal p_S of the coordinate subspace of the axes S as far as it can, the side B does the same along p_T , and the remainders a' , b' are placed by Carver's criterion in the plane of the axes k, l . The following special case of (II) (see Section 7) needs no plane at all:

(I) there are disjoint $S, T \subseteq [n]$ with $A \leq \|p_S\|$ and $B \leq \|p_T\|$.

Example 2.2. The square $A = B$ in the unit cube $p = (1, 1, 1)$. Take $(k, l) = (1, 2)$, $S = \{3\}$, $T = \emptyset$. Then $a' = \sqrt{A^2 - 1}$ and $b' = A$, and for $A > 1$ Carver's condition (ii) in the unit square is $(2/(A + \sqrt{A^2 - 1}))^2 \geq 2$, that is, $A + \sqrt{A^2 - 1} \leq \sqrt{2}$, or $A \leq 3\sqrt{2}/4 \approx 1.0607$. The other checks give nothing better, so this is the largest square in the unit cube, in accordance with Nieuwland's result. Figure 1 shows the position given by Section 3: the vertices are $(\frac{1}{4}, 0, 0)$, $(0, \frac{1}{4}, 1)$, $(\frac{3}{4}, 1, 1)$, $(1, \frac{3}{4}, 0)$; the side v lies in the plane of the axes 1, 2, and the side u has the component $A u_3 = 1 = p_3$ along the axis $3 \in S$.

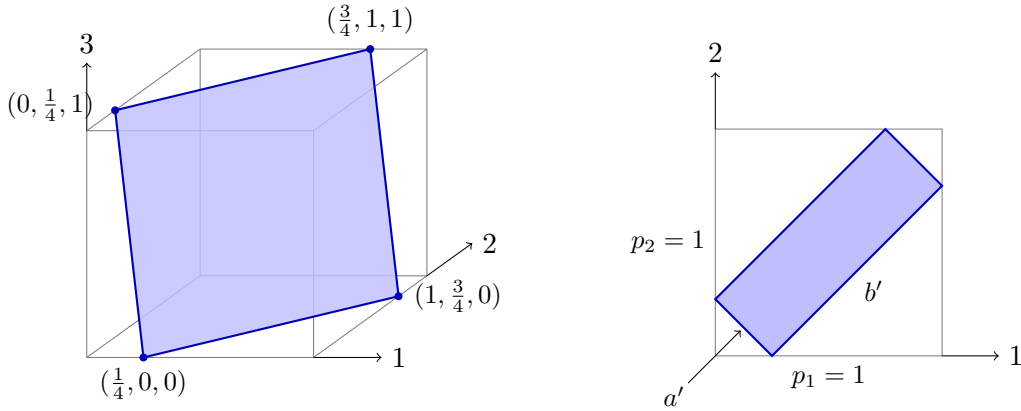


Figure 1: Left: the largest square $A = B = 3\sqrt{2}/4$ in the unit cube, in the position constructed in Section 3 for $(k, l) = (1, 2)$, $S = \{3\}$, $T = \emptyset$. Right: its projection to the plane of the axes 1, 2 is the $a' \times b'$ rectangle, $a' = \sqrt{A^2 - 1} = \sqrt{2}/4$, placed at 45° by Carver's criterion; the rest of the side A goes along the axis 3.

3 Sufficiency

Assume (II). Carver's criterion gives orthonormal $f_a, f_b \in \text{span}(e_k, e_l)$ with $a'|(f_a)_i| + b'|(f_b)_i| \leq p_i$ for $i \in \{k, l\}$. Let $\lambda = \min(1, A/\|p_S\|)$ and $\mu = \min(1, B/\|p_T\|)$ (with $\lambda = 0$ if $\|p_S\| = 0$ and $\mu = 0$ if $\|p_T\| = 0$), $p_S = \sum_{i \in S} p_i e_i$, and

$$A u = \lambda p_S + a' f_a, \quad B v = \mu p_T + b' f_b.$$

- Norms: $\|A u\|^2 = \lambda^2 \|p_S\|^2 + a'^2 = A^2$ both for $A \geq \|p_S\|$ and for $A < \|p_S\|$ (then $a' = 0$). Similarly $\|B v\| = B$.
- Orthogonality: the supports S , T and $\{k, l\}$ are pairwise disjoint and $f_a \perp f_b$, so $u \perp v$.
- Widths: on S we have $A|u_i| = \lambda p_i \leq p_i$ and $v_i = 0$; on T symmetrically; on $\{k, l\}$ by Carver's criterion.

By Section 4.1 this is a position of the rectangle in the box. (If $A = 0$ put $u = f_a$, and if $B = 0$ put $v = f_b$.) $\square$

4 Reformulation

4.1 Reduction to widths

A box is the intersection of the slabs $\{0 \leq x_i \leq p_i\}$, and shifts along different coordinates are independent. Hence a convex body can be translated into P if and only if its width along every e_i is at most p_i . The width of the rectangle (Au, Bv) along e_i is $A|u_i| + B|v_i|$. Thus

$$\text{the rectangle fits} \iff \exists \text{ orthonormal } u, v : \quad A|u_i| + B|v_i| \leq p_i \quad \text{for all } i. \quad (1)$$

The case $B = 0$ (a segment): it fits if and only if $A \leq \|p\|$; the necessity follows from $A^2 = \sum A^2 u_i^2 \leq \sum p_i^2$. This is (I) with $S = [n]$, $T = \emptyset$. From now on $A, B > 0$. Axes with $p_i = 0$ are discarded at once: on them $u_i = v_i = 0$. If S, T, k, l satisfying (I) or (II) are found for the remaining axes, each discarded axis is added to S ; this only increases $\|p_S\|$, hence decreases a' and keeps Carver's condition, and it keeps $A \leq \|p_S\|$ in (I).

If fewer than two axes have $p_i > 0$, then for $A, B > 0$ the rectangle does not fit (two orthonormal vectors cannot be placed into one coordinate), and (II) is false as well. Indeed, let m be the only axis with $p_m > 0$ (or there is none). If $m \in \{k, l\}$, then $a' = A > 0$ and $b' = B > 0$, while the box $(p_m, 0)$ holds only a segment; if $m \in S$, then $b' = B > 0$ and the box $(0, 0)$ is a point; if $m \in T$ (or there is no m), then $a' = A > 0$ and again the box is a point. So the theorem holds in this case too.

4.2 Relaxation

In (1) one may require only $\|u\| \geq 1$, $\|v\| \geq 1$, $\langle u, v \rangle = 0$: shrinking u or v by a factor ≤ 1 keeps the orthogonality and the inequalities. Write $P_i = p_i^2$, $\alpha_i = A|u_i|/p_i$, $\beta_i = B|v_i|/p_i$ ($\alpha_i + \beta_i \leq 1$) and $\sigma_i = \text{sgn}(u_i v_i) \in \{\pm 1\}$ (any sign if $u_i v_i = 0$). The conditions read

$$\sum P_i \alpha_i^2 \geq A^2, \quad \sum P_i \beta_i^2 \geq B^2, \quad \sum \sigma_i P_i \alpha_i \beta_i = 0.$$

4.3 Active axes

Let $\rho_i = \alpha_i + \beta_i \in [0, 1]$ and $p'_i = \rho_i p_i = A|u_i| + B|v_i| \leq p_i$. In the box p' with the same u, v every axis is *active*: $\alpha'_i + \beta'_i = 1$. Put $\alpha'_i = (1 + t_i)/2$, $\beta'_i = (1 - t_i)/2$, $t_i \in [-1, 1]$. The conditions of Section 4.2 for p' are the same, since $P'_i \alpha_i'^2 = A^2 u_i^2$ and so on. Axes with $\rho_i = 0$ are discarded.

Conditions (I) and (II) are monotone in p : when p grows, the norms $\|p_S\|$ grow, so a', b' do not grow, while the box (p_k, p_l) grows. Hence (I) $\vee$ (II) for p' implies (I) $\vee$ (II) for p . *Therefore it suffices to prove that an active configuration in the box p satisfies (I) or (II).* From now on we keep the notation p for the box in which all axes are active.

An active configuration is a pair (t, σ) , $t \in [-1, 1]^n$, $\sigma \in \{\pm 1\}^n$, with $A|u_i| = p_i(1 + t_i)/2$, $B|v_i| = p_i(1 - t_i)/2$ and $\text{sgn}(u_i v_i) = \sigma_i$. The *sides* are $+$ $= \{i : \sigma_i = +1\}$ and $-$ $= \{i : \sigma_i = -1\}$. An axis is *mixed* if $|t_i| < 1$, that is, $u_i v_i \neq 0$.

Conversely, let all $p_i > 0$ and let (t, σ) be any pair. Put $|u_i| = p_i(1 + t_i)/(2A)$, $|v_i| = p_i(1 - t_i)/(2B)$ and choose the signs so that $\text{sgn}(u_i v_i) = \sigma_i$ whenever $u_i v_i \neq 0$. Then all widths equal p_i , and

$$\|u\|^2 = \sum \frac{P_i(1 + t_i)^2}{4A^2}, \quad \|v\|^2 = \sum \frac{P_i(1 - t_i)^2}{4B^2}, \quad \langle u, v \rangle = \sum \frac{\sigma_i P_i(1 - t_i^2)}{4AB}.$$

We call (t, σ) *admissible* if $\|u\| \geq 1$, $\|v\| \geq 1$ and $\langle u, v \rangle = 0$. By Section 4.2 an admissible configuration gives a position of the rectangle. All configurations built in Section 5 are given by pairs (t', σ') , and their admissibility is checked through these three conditions only; an equivalent form of them follows.

4.4 The tent

Let $L = \sum P_i t_i$, $\text{Def}_\pm = \sum_{i \in \pm} P_i(1 - t_i^2)$, $W = \sum P_i$, $c_A = 2A^2 - W$, $c_B = W - 2B^2$.

- Orthogonality $\sum \sigma_i P_i(1 - t_i^2) = 0$ means $\text{Def}_+ = \text{Def}_- =: D$.
- Then $\sum P_i t_i^2 = W - 2D$, and $\sum P_i(1 + t_i)^2 \geq 4A^2 \iff L - D \geq c_A$, $\sum P_i(1 - t_i)^2 \geq 4B^2 \iff L + D \leq c_B$.

So a configuration with $\text{Def}_+ = \text{Def}_- = D$ is admissible if and only if the point (L, D) lies under the “tent” $\{D \leq L - c_A, D \leq c_B - L\}$ (Figure 2). The tent is 1-Lipschitz, hence:

($\star$) If (t, σ) is admissible and (t', σ') is another configuration with $\text{Def}'_+ = \text{Def}'_- = D'$ and $D' + |L' - L| \leq D$, then (t', σ') is admissible. Indeed, $L' - D' \geq L' - D + |L' - L| \geq L - D \geq c_A$ and $L' + D' \leq L + D \leq c_B$.

A mixed axis contributes a positive amount to Def of its side. Since $\text{Def}_+ = \text{Def}_-$, mixed axes are either present on both sides or absent altogether; exactly one mixed axis is impossible.

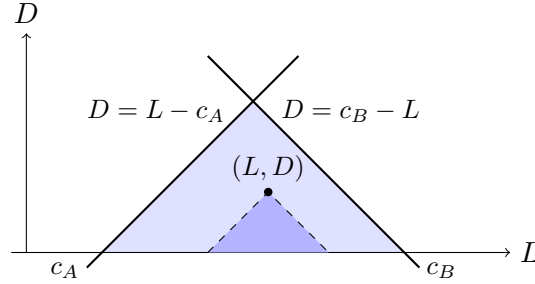


Figure 2: Admissible configurations correspond to points (L, D) under the tent. Every point (L', D') with $D' + |L' - L| \leq D$ (the darker triangle below (L, D)) is again under the tent; this is ($\star$).

5 Three lemmas

Lemma 5.1 (C: at most two mixed axes). *If an admissible configuration has no mixed axes, then (I) holds. If it has exactly two mixed axes, $k \in +$ and $l \in -$, then (II) holds with these k, l .*

Proof. No mixed axes. Put $S = \{t = +1\}$, $T = \{t = -1\}$. Then $D = 0$, $L = \|p_S\|^2 - \|p_T\|^2$, $W = \|p_S\|^2 + \|p_T\|^2$, and the tent gives $\|p_S\|^2 \geq A^2$, $\|p_T\|^2 \geq B^2$; this is (I).

Two mixed axes $k \in +$, $l \in -$. On the other axes $u_i v_i = 0$, so for $U = (u_k, u_l)$, $V = (v_k, v_l)$ we have $\langle U, V \rangle = \langle u, v \rangle = 0$; also $U, V \neq 0$ because $u_i v_i \neq 0$ on a mixed axis. Hence the rectangle $A|U| \times B|V|$ with directions $U/|U|$, $V/|V|$ lies in the 2-box (p_k, p_l) : its widths are $A|u_i| + B|v_i| = p_i$. Let $S = \{t = +1\}$; there $v_i = 0$ and $A|u_i| = p_i$. Then $A^2|U|^2 = A^2\|u\|^2 - \|p_S\|^2 \geq A^2 - \|p_S\|^2$ (the axes with $t = -1$ do not contribute to u), that is, $A|U| \geq a'$. Similarly $B|V| \geq b'$ for $T = \{t = -1\}$. By monotonicity $\text{Carver}(a', b'; p_k, p_l)$ holds; this is (II). $\square$

Lemma 5.2 (R: opposite signs). *Suppose all mixed axes of the side $+$ have $t \geq 0$ and all mixed axes of the side $-$ have $t \leq 0$ (or the other way round). Then (I) holds.*

Proof. Round: $t \mapsto +1$ on the mixed axes of the side $+$ and $t \mapsto -1$ on the mixed axes of the side $-$.

- The shift of L from the side $+$ is $\Delta_+ = \sum P(1-t) \in [0, D]$, since $1-t \leq 1-t^2$ for $t \in [0, 1]$.
- The shift of L from the side $-$ is $\Delta_- = -\sum P(1+t) \in [-D, 0]$, since $1+t \leq 1-t^2$ for $t \in [-1, 0]$.
- In the new configuration $\text{Def}'_+ = \text{Def}'_- = 0$ and $|L' - L| = |\Delta_+ + \Delta_-| \leq D$, the summands having opposite signs.

By $(\star)$ it is admissible. It has no mixed axes, so Lemma 5.1 gives (I). The other case is symmetric. $\square$

Lemma 5.3 (V: moving along a curve). *Let $i \neq j$ be mixed axes of one side (say $+$) and k a mixed axis of the other side, $s_0 = t_k$. Suppose that it is not the case that $t_i = t_j$ and $t_i s_0 \leq 0$; that is, either (a) $t_i \neq t_j$, or (b) $t_i = t_j$ and $t_i s_0 > 0$. Then there is an admissible configuration with a smaller D .*

Proof. We change only (t_i, t_j, t_k) , keeping L and $\text{Def}_+ - \text{Def}_-$:

$$P_i t_i + P_j t_j = \lambda(s) := L_0 - P_k s, \quad P_i t_i^2 + P_j t_j^2 = \mu(s) := K_0 + P_k s^2, \quad t_k = s.$$

Then $D = \text{Def}_- = \text{const} + P_k(1-s^2)$ strictly decreases as $|s|$ grows, while L does not change. By $(\star)$ the configuration stays admissible as long as $(t_i, t_j, t_k) \in [-1, 1]^3$. (The other mixed axes of the side $+$, if any, enter the constants L_0, K_0 .)

The line $\{P_i X + P_j Y = \lambda\}$ and the ellipse $\{P_i X^2 + P_j Y^2 = \mu\}$ intersect if and only if $h(s) := \mu(s) - \lambda(s)^2/(P_i + P_j) \geq 0$: by the Cauchy–Schwarz inequality the minimum of $P_i X^2 + P_j Y^2$ on the line is attained at $X = Y = \tau := \lambda/(P_i + P_j)$, and $h = 0$ means tangency, $t_i = t_j = \tau$. We have $h'(s) = 2P_k(s + \tau)$.

(a) If $t_i \neq t_j$, the Jacobian of $(X, Y) \mapsto (P_i X + P_j Y, P_i X^2 + P_j Y^2)$ equals $2P_i P_j(Y - X) \neq 0$, and by the implicit function theorem the solution extends to s close to s_0 , in particular to $|s| > |s_0|$ (for $s_0 = 0$ in either direction). The point (t_i, t_j) is inside the square $(-1, 1)^2$ and $|s_0| < 1$, so for a small move everything stays in $[-1, 1]$, and D strictly decreases.

(b) If $t_i = t_j = \tau$ and $\tau s_0 > 0$, then $s_0 \neq 0$, $h(s_0) = 0$, and $h'(s_0) = 2P_k(s_0 + \tau)$ has the sign of s_0 . Hence $h > 0$ for a small increase of $|s|$, and there are two intersection points. Since i, j are mixed, $|\tau| < 1$; choose $\varepsilon > 0$ with $|\tau| + 2\varepsilon < 1$. Both intersection points depend continuously on s and tend to (τ, τ) as $s \rightarrow s_0$ (the distance to the tangency point is of order $\sqrt{h}$). So for a small enough change of s both coordinates stay in $(-|\tau| - \varepsilon, |\tau| + \varepsilon) \subset (-1, 1)$, and also $|s| < 1$. Again D strictly decreases. $\square$

6 Necessity

By Section 4 it suffices to consider an admissible active configuration in the box p with all $p_i > 0$. For each of the finitely many σ the set of admissible $t \in [-1, 1]^n$ is given by the closed conditions $\text{Def}_+ = \text{Def}_-$, $L - D \geq c_A$, $L + D \leq c_B$, where L , $\text{Def}_\pm$ and D are polynomials in t ; so it is compact. For the σ of the given configuration it is nonempty, and $D \geq 0$ is continuous, so there is a configuration with the *smallest* D .

1. If some side has two mixed axes with different t , then the other side has a mixed axis too (Section 4.4), and Lemma 5.3(a) decreases D , a contradiction. So all mixed axes of the side $+$ have a common $t = \tau$, and all mixed axes of the side $-$ a common $t = s$.
2. If each side has at most one mixed axis, then by Section 4.4 there are 0 or $1 + 1$ of them, and Lemma 5.1 applies.

3. Otherwise some side (say $+$) has at least two mixed axes with the common τ . Take a mixed axis k of the other side, $t_k = s$. If $\tau s > 0$, Lemma 5.3(b) decreases D , a contradiction. So $\tau s \leq 0$, the sign conditions of Lemma 5.2 hold, and we get (I). $\square$

Corollary 6.1. *Suppose (I) fails. Then every admissible configuration with the smallest D has exactly two mixed axes, one on each side, and their t have the same sign. In other words, an optimal position has the form (II) with a rotation in a single coordinate plane.*

Proof. No mixed axes would give (I) by Lemma 5.1; two or more on one side lead by items 1 and 3 to a contradiction or to (I); opposite signs would give (I) by Lemma 5.2. $\square$

7 (I) is a special case of (II)

Let $A, B > 0$ and (I) hold. Then S and T are nonempty, since $\|p_S\| \geq A > 0$ and $\|p_T\| \geq B > 0$. Let $R = [n] \setminus (S \cup T)$, take $k \in S$, $l \in T$ and put $S' = (S \setminus \{k\}) \cup R$, $T' = T \setminus \{l\}$, so that $S' \sqcup T' = [n] \setminus \{k, l\}$. Then $\|p_{S'}\|^2 = \|p_S\|^2 - p_k^2 + \|p_R\|^2 \geq A^2 - p_k^2$, whence $a'^2 \leq p_k^2$; similarly $b'^2 \leq p_l^2$, and Carver's condition holds without rotation. For $B = 0$ take any $k \neq l$, $S =$ the other axes, $T = \emptyset$: the segment a' fits into (p_k, p_l) if and only if $a' \leq \sqrt{p_k^2 + p_l^2}$, that is, $A \leq \|p\|$.

Together with Sections 3–6 this proves Theorem 2.1. As by-products: (1) has a solution with at most two coordinates where $u_i v_i \neq 0$; and if (I) fails, the two mixed axes in (II) have t of the same sign.

8 The unit cube: Theorem 5 of Jerrard and Wetzel

Jerrard and Wetzel [2, Theorem 5, formula (8)] found the largest $L \times \lambda L$ rectangle, $0 \leq \lambda \leq 1$, in the unit cube: $L_{\max} = f_1(\lambda)$ on $[0, \lambda_1]$, $\sqrt{2}$ on $[\lambda_1, \lambda_2]$ and $f_3(\lambda)$ on $[\lambda_2, 1]$, where

$$f_1(\lambda) = \frac{3}{\sqrt{3 - \lambda^2} + \sqrt{2}\lambda}, \quad f_3(\lambda) = \frac{3}{\sqrt{2} + \sqrt{3\lambda^2 - 1}}, \quad \lambda_1 = 1 - \frac{\sqrt{2}}{2}, \quad \lambda_2 = \frac{\sqrt{2}}{2}.$$

Theorem 2.1 gives the same answer by a different route. Let $p = (1, 1, 1)$, $A = L$, $B = \lambda L$, and let m be the third axis, $\{k, l, m\} = \{1, 2, 3\}$.

- $m \in S$, $A > 1$: $a' = \sqrt{A^2 - 1}$, $b' = \lambda A$, and Carver in the unit square gives $\sqrt{A^2 - 1} + \lambda A \leq \sqrt{2}$, that is, $(1 - \lambda^2)A^2 + 2\sqrt{2}\lambda A - 3 \leq 0$, or $A \leq f_1(\lambda)$.
- $m \in T$, $B \leq 1$: $b' = 0$, a segment A in the unit square, $A \leq \sqrt{2}$; here $B = \lambda\sqrt{2} \leq 1$ means $\lambda \leq \sqrt{2}/2$.
- $m \in T$, $B > 1$: $a' = A$, $b' = \sqrt{\lambda^2 A^2 - 1}$, and Carver gives $A + \sqrt{\lambda^2 A^2 - 1} \leq \sqrt{2}$, that is, $A \leq f_3(\lambda)$. This is feasible only for $\lambda \geq \lambda_2$: the left-hand side grows in A and equals $1/\lambda$ at $A = 1/\lambda$, so $1/\lambda \leq \sqrt{2}$ is needed.
- The rectangle without rotation gives $\min(\sqrt{2}, 1/\lambda)$, which is never better.

The denominators of f_1 and f_3 increase in λ , so f_1 and f_3 decrease, and $f_1(\lambda_1) = f_3(\lambda_2) = \sqrt{2}$. On $[\lambda_2, 1]$ we have $f_3 \geq f_1$: the difference of the denominators $\varphi = (\sqrt{3 - \lambda^2} + \sqrt{2}\lambda) - (\sqrt{2} + \sqrt{3\lambda^2 - 1})$ vanishes at $\lambda = 1$ and decreases, since $\varphi' \leq \sqrt{2} - 3\lambda/\sqrt{3\lambda^2 - 1} \leq \sqrt{2} - 3/\sqrt{2} < 0$. So the maximum over these variants is exactly $L_{\max}(\lambda)$. Jerrard and Wetzel argue by the positions of the vertices; here the variants are partitions of the axes. Numerically, the enumeration of all (k, l, S, T) with bisection agrees with formula (8) on 1001 values of $\lambda \in [0, 1]$ up to $3.3 \cdot 10^{-16}$.

9 The formal proof

The theorem has been proved in Lean 4 with Mathlib; the sources are in the directory `lean/` of the repository [4]. The meaning of the main theorem depends only on the following definitions (file `Widths.lean`):

```
abbrev En (n : ℕ) := EuclideanSpace ℝ (Fin n)
def boxN (q : Fin n → ℝ) : Set (En n) := {x | ∀ i, 0 ≤ x i ∧ x i ≤ q i}
noncomputable def pad (h : m ≤ n) (q : Fin m → ℝ) : Fin n → ℝ :=
  Function.extend (Fin.castLE h) q 0
def FitsN (p q : Fin n → ℝ) : Prop := ∃ f : En n ≃ En n, f '' boxN q ⊆ boxN p
```

Here `pad h ! [A, B]` is the vector of edges $(A, B, 0, \dots, 0)$, so the rectangle is the box `boxN (pad h ! [A, B])` in $\mathbb{R}^n$, and `FitsN` asks for an isometry of $\mathbb{R}^n$ (any rigid motion, reflections included) mapping one box into the other. Carver’s criterion is written verbatim as in Section 2 (`CarverS` for sorted sides, `Carver` sorts by `max/min`). The main theorem (file `RectCarver.lean`) reads

```
theorem fitsN_rect_iff_explicit (h : 2 ≤ n) (p : Fin n → ℝ) (hp : ∀ i, 0 ≤ p i)
  {A B : ℝ} (hA : 0 ≤ A) (hB : 0 ≤ B) :
  FitsN p (pad h ! [A, B]) ↔
  ∃ k l : Fin n, k ≠ l ∧ ∃ S T : Finset (Fin n), Disjoint S T ∧
    S ∪ T = univ \ {k, l} ∧
    Carver (√(A ^ 2 - ∑ i ∈ S, p i ^ 2)) (√(B ^ 2 - ∑ i ∈ T, p i ^ 2))
      (p k) (p l)
```

(in Mathlib $\sqrt{x} = 0$ for $x \leq 0$, so the positive part is automatic). The command `#print axioms` reports only the standard axioms `propext`, `Classical.choice`, `Quot.sound`; the sources contain no `sorry`. Versions: Lean 4 v4.34.0-rc2, Mathlib v4.34.0-rc2. The hypotheses are only $n \geq 2$, $p_i \geq 0$, $A, B \geq 0$, so the degenerate cases $A = 0$ or $B = 0$ (a segment, a point) are covered by the same statement.

The formalization follows Sections 3–7 with a few simplifications. Instead of passing to active axes, it works with $x = Au$, $y = Bv$ and the local widths $r_i = |x_i| + |y_i| \leq p_i$, which all steps preserve; it minimizes $\sum |x_i y_i|$ (which is $D/2$) over all solutions of the relaxed problem at once, without fixing σ ; and Lemma 5.3 is proved without the implicit function theorem, by the explicit parametrization $X = \tau + P_j \theta$, $Y = \tau - P_i \theta$ of the line, for which $P_i X^2 + P_j Y^2 = (P_i + P_j) \tau^2 + P_i P_j (P_i + P_j) \theta^2$. Carver’s criterion itself is proved from the strip lemma of [3]: if $a > X$, the optimal position of the $a \times b$ rectangle touches the strip of width X , and it remains to compare its extent along the strip with Y , which is pure algebra.

10 Computation

The criterion runs through $\frac{1}{2}n(n-1)2^{n-2}$ cases, each a Carver test with two square roots; the number of cases grows exponentially with n , but for $n \leq 10$ it is at most 11 520. Further, Section 3 turns a successful case into an explicit orthonormal pair (u, v) . We compared a Python implementation (`tools/rect_fit.py` in [4]) with two numerical searches over positions: the Nelder–Mead method on $F(u, v) = \max_i (A|u_i| + B|v_i|)/p_i$ over orthonormal pairs (as in [3]), and the SLSQP method on the smooth lifted problem “minimize t subject to $Aa_i + Bb_i \leq tp_i$, $a_i \geq |u_i|$, $b_i \geq |v_i|$, $\|u\| = \|v\| = 1$, $u \perp v$ ”, both from random starts. A numerical search answers “fits” as soon as some start reaches $F \leq 1$; it cannot confirm that the rectangle does not fit. For each n we took 50 pairs near the

boundary: the rectangle is scaled by $t^*(1 \pm e)$, where t^* is the largest scale at which it fits and $10^{-4} \leq e \leq 10^{-2}$; about half of them fit. Timings per pair (Python 3.10, one core). The criterion stops at the first successful case; the column “criterion” is averaged over all pairs, including those that do not fit and need all cases, while the column “criterion + position” is averaged over the fitting pairs only, which is why it can be smaller.

n	criterion	criterion + position	Nelder–Mead, 20 starts	SLSQP, 20 starts
3	$6 \mu\text{s}$	$7 \mu\text{s}$	1.0 s; missed 3 of 25	26 ms; missed 1 of 25
4	$26 \mu\text{s}$	$21 \mu\text{s}$	1.9 s; missed 19 of 22	40 ms; missed 1 of 22
5	$80 \mu\text{s}$	$36 \mu\text{s}$	2.4 s; missed 24 of 24	46 ms; missed 2 of 24
6	$194 \mu\text{s}$	$42 \mu\text{s}$	3.3 s; missed 24 of 24	63 ms; missed 3 of 24

“Missed” means that the search answered “does not fit” for a rectangle that fits. No search ever found a position where the criterion says “does not fit”. All positions produced by the criterion passed the check (orthonormality and widths), both here and in a separate self-test of the implementation on 20 000 random pairs with $n = 2, \dots, 7$ (not timed). Floating-point comparisons are exact only in exact arithmetic: near the boundary rounding may flip the answer.

11 Remarks

The argument uses a single pair of orthogonal columns (u, v) , that is, a single orthogonality condition, which takes the form $\text{Def}_+ = \text{Def}_-$. This is why the problem fits into the two-dimensional picture (L, D) of the tent. For a box with three nonzero edges there are three such conditions; each pair of the columns gives its own partition of the axes into sides, and a move of Lemma 5.3 preserving one condition breaks the other two. This is an explanation, not a proof. For a three-dimensional box in a three-dimensional box an explicit criterion is known [3]; for $k \geq 3$ and $d \geq 4$ the problem of Jerrard and Wetzel remains open, as far as we know.

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